Ishaan Gandhi

ishaangandhi@gmail.com | github.com/ishaangandhi | [US Citizen]

Experience

ElevenLabs Apr. 2025 - Present
Senior Research Engineer *New York, NY*

- Developed data pipelines, evaluations, and environments for text-to-speech post-training and speech-to-text.
- Fine-tuned encoder-only and decoder-only open-weight models for research purposes.
- Coordinated with research engineers, researchers, and data labellers.

Uben AI Mar. 2024 - Mar. 2025
Founder and CEO *New York, NY*

- Founded and created Uben AI, a company that uses AI to help people learn languages and pass residency exams.
- Fine-tuned LLAMA 3.1 using LoRA for company use-cases.
- Raised capital, hired and managed a team of developers, designers, and marketers. Sold product to hundreds of paying subscribers.

Jane Street Apr. 2023 - Mar. 2024
Software Engineering - Fixed Income *New York, NY*

- Used PyTorch to create algorithms for pricing interest rate swaps and construct interest rate curves.
- Profiled optimization, conducted hyperparameter tuning, optimized performance and memory usage for these algorithms.

Citadel Securities Aug. 2021 - Feb. 2023
Software Engineering - Developer Infrastructure *New York, NY*

- Found and fixed bugs in various LLVM projects, including clang-tidy and clang-format.
- Improved code quality across the firm by integrating clang-tidy into the code review tool, which discovers and fixes hundreds of would-be bugs every day.
- Added support for code coverage, which was previously nearly impossible to collect, in the build system, CI system, and code review tool. This is now used by most developers every day, and has lead to the discovery of latent bugs.

Facebook Summer 2019
Software Engineering Intern - Place Visit Detection *New York, NY*

- Implemented 2-pass scoring in Facebook's machine learning model for place visit detection, increasing model AUC by 0.74pp.
- Performed performance optimization to improve speed and reduce memory usage of new model. Worked on prediction service and training workflow in C++ and Python respectively.

Facebook Summer 2018
Software Engineering Intern - Real time Infrastructure *Seattle, WA*

- Worked on Facebook's pub-sub system (C++ and Java) on the real-time infrastructure team. Improved the security of infrastructure by using cryptographic authentication tokens to authenticate subscribe requests to certain topics.

Open Source Contributions

- Wrote, tested, and submitted an implementation of RFC 5837 (Extending ICMP for Interface and Next-Hop Identification) for **Linux 5.12**.
- Fixed various bugs in **LLVM**, including within the C++ language server **clangd** and the C++ linter **clang-tidy**.
- Improved static-analysis in **LLVM's** convergent attributor, which enables more opportunities for GPU-acceleration.
- Enhanced the **tracepath** and **traceroute6** applications in the widely used iputils package to identify arrival and next-hop interfaces when such information is present in ICMP replies.
- Added support for the IPv6 compact routing header (CRH), and IPv6 tunnel payload forwarding (TPF) in **Wireshark**. Also fixed six bugs related to ICMP extensions.

- Added a new dissector for the RFC 5837 protocol in **TCPDump**.
- Fixed bugs and added interactive parsing mode to the POSIX compliant shell parsing libraries **Morbig** and **Morsmall**.

Projects

GPU-Accelerated Animations | *CUDA*
- Wrote CUDA kernels to create graphics and animations of overlapping textures, including falling snow.

LLVM Interpolation Pass | *C++*
- Wrote a compiler optimization for LLVM to make arrays easier to understand for software verification tools.
- The optimization converted lookups into arrays whose contents were known at compile time into function calls that could be more easily understood by verification tools.

Symbolic Executor | *Python*
- Hired by Harvey Mudd's CS department to design a project for the *Applied Logic* course. Wrote a symbolic executor for a subset of Python.

TCP/IP Stack | *C++*
- Wrote an implementation of a networking stack in using modern C++ (C++17). Included the transmission control protocol (TCP), address resolution protocol (ARP), and IP router.
- Profiled with Gprof and improved throughput of implementation 0.05 Gbit/s to 0.78 Gbit/s.
- Used this stack to talk to real internet servers in a command line application similar to cURL.

Publications

- **Ishaan Gandhi**, Anshula Gandhi. Lightening the Cognitive Load of Shell Programming. 11th Annual Workshop on Evaluation and Usability of Programming Languages and Tools (PLATEAU 2020.)

Research

ETH Zürich | *Advanced Software Technology Lab* Spring 2020
- Used a coverage-guided fuzzer (AFL) to find bugs in the SMT solvers CVC4 and Z3. Profiled the symbolic executor KLEE using C++ and Python.

Harvey Mudd College | *Parallel Symbolic Execution Research* Spring 2021
- Analyzing novel matrix-based approach to symbolic execution. Wrote tool to generate control-flow-graphs for a proprietary language and convert these CFGs into matrices.

Pomona College | *Formalization of the POSIX shell standard* Fall 2020
- Worked on integrating the Morbig parser to the SMOOSH shell in OCaml.
- Found and fixed bugs in the **Morbig** and **Morsmall** parser libraries, and extended them to support interactive parsing.

Technical Skills

C, C++ (up to standards in C++17), OCaml, Coq, Haskell, Java, Go, Swift, SQL, Bash, Git, Flask, JS, React, AWS, Kafka, Mongo

Education

Harvey Mudd College Claremont, CA
Bachelor of Science in Mathematics & Computer Science *2017 – 2021*

Graduated with departmental honors and a 3.9 GPA as a *Harvey S. Mudd scholar*. Selected coursework: Computer Systems, Databases, Operating Systems, Programming Languages, Software Verification, Advanced Algorithms, Algorithmic Game Theory, Model Theory.

Test Scores

ACT: (99.9 percentile)

Math:	36/36
English:	36/36
Reading:	36/36
Science:	36/36